LECTURE 29

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$$V = \int_a^b \int_{g_1(x)}^{g_2(x)} \int_{f_1(x,y)}^{f_2(x,y)} 1 \, dz \, dy \, dx$$

$$\int_x^y \int_1^2 \int_0^1 \quad \times$$

6 -forms

$$\begin{cases} dz\ dy\ dx \\ dz\ dx\ dy \\ dy\ dx\ dz \\ dy\ dz\ dx \\ dx\ dy\ dz \\ dx\ dz\ dy \end{cases}$$

1) $\iiint_D xyz \, dx \, dy \, dz$

$D = [0,1] \times [1,2] \times [2,3]$

$$= \int_2^3 \int_1^2 \int_0^1 xyz \, dx \, dy \, dz$$

$$= \int_2^3 \int_1^2 yz \left(\int_0^1 x \, dx \right) dy \, dz$$

$$= \int_2^3 \int_1^2 yz \left[\frac{x^2}{2} \right]_0^1 dy \, dz$$

$$= \frac{1}{2} \int_2^3 \int_1^2 yz \, dy \, dz$$

$$= \frac{1}{2} \int_2^3 z \left(\int_1^2 y \, dy \right) dz$$

$$= \frac{1}{2} \int_2^3 z \left[\frac{y^2}{2} \right]_1^2 dz = \frac{1}{2} \cdot \frac{3}{2} \int_2^3 z \, dz$$

$$= \frac{1}{2} \int_2^3 z \left[\frac{y^2}{2} \right]_1^2 dz$$

$$= \frac{3}{4} \left[\frac{z^2}{2} \right]_2^3 = \frac{15}{8}$$

(2) Volume of space bdd by $x + 2y + 3z = 4$ in 1st quadrant

$$\int_{x=0}^{x=4} \int_{y=0}^{y=\frac{4-x}{2}} \int_{z=0}^{z=\frac{4-x-2y}{3}} dz \, dy \, dx$$

$x + 2y = 4$

$y = \frac{4-x}{2}$

$$\int_{x=0}^{4} \int_{y=0}^{\frac{4-x}{2}} \left[z \right]_{z=0}^{\frac{4-x-2y}{3}} dy \, dx$$

$x=0 \quad y=0 \quad z=0$

$$\int_0^4 \int_0^{\frac{4-x}{2}} [z]_0^{\frac{4-x-2y}{3}} \, dy \, dx$$

$$= \int_0^4 \int_0^{\frac{4-x}{2}} \frac{4-x-2y}{3} \, dy \, dx$$

$$= \frac{1}{3} \int_0^4 \int_0^{\frac{4-x}{2}} (4-x-2y) \, dy \, dx$$

$$= \frac{1}{3} \int_0^4 \left[4y - xy - y^2 \right]_0^{\frac{4-x}{2}} dx = \frac{16}{9}$$

3) volume of region bdd by $x^2+y^2=1$ & $z=0, z=-y$

$$\int_{-1}^{1} \int_{-\sqrt{1-x^2}}^{0} \int_0^{-y} dz \, dy \, dx \qquad y = \pm\sqrt{1-x^2}$$

$$= \frac{2}{3} \text{ units}$$ H.W.

Cylindrical co-ordinate System : (r, θ, z)

$x = r\cos\theta, \quad y = r\sin\theta, \quad z = z \qquad \underset{(r,\theta,z)}{C} \to \underset{(x,y,z)}{R}$

Rectangular $\to$ Cylindrical $\quad r = \sqrt{x^2+y^2} \quad \tan\theta = \frac{y}{x}, \quad z = z$

$(x, y, z) \qquad (r, \theta, z)$

$R: (3, \sqrt{3}, 2) \to C$

$x \quad y \quad z \qquad (r, \theta, z) \qquad r = \sqrt{12}, \ \theta = \frac{\pi}{6}, \ z = 2$

$C: \left(2, \frac{-\pi}{4}, 1\right) \to R$

$r, \theta, z \to x = \quad y = \quad z = 1$ H.W.

Spherical co ordinate system (ρ, θ, ϕ)

$S \to R \quad x = \rho\sin\phi\cos\theta, \quad y = \rho\sin\phi\sin\theta \qquad z = \rho\cos\phi$

$R \to S \qquad \rho = \sqrt{x^2+y^2+z^2} \qquad \tan\theta = \frac{y}{x} \qquad \cos\phi = \frac{z}{\sqrt{\quad}}$

$R \to S \qquad \rho = \sqrt{x^2+y^2+z^2} \qquad \tan\theta = \frac{y}{x} \qquad \cos\phi = \frac{z}{\sqrt{x^2+y^2+z^2}}$

$= \frac{z}{\rho}$

$R(3, 3\sqrt{3}, 2) \longrightarrow S(\rho, \theta, \phi)$

$4, \frac{\pi}{6}, \frac{\pi}{3}$

$S \left(1, \frac{\pi}{2}, \frac{\pi}{4}\right) \to R(x, y, z)$

ρ, θ, ϕ

$\left(0, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$